Copilot

Certainly! Let's talk about **prefix sums** in Java. Prefix sums (also known as cumulative sums) are a powerful technique used in competitive programming and algorithmic problem-solving. They allow us to efficiently compute the sum of elements in a given range of an array.

What is a Prefix Sum?

A prefix sum (or cumulative sum) array is an auxiliary array where each element represents the sum of all elements up to that point in the original array. It's a handy precomputation that can significantly speed up certain types of queries.

Implementation in Java

Let's start with a simple example. Suppose we have an array `arr` of integers:

```
int[] arr = {10, 4, 16, 20};
```

We want to compute the prefix sum array for this input. Here's how you can do it in Java:

```
public class PrefixSumExample {
    public static void fillPrefixSum(int[] arr, int n, int[] prefixSum) {
        prefixSum[0] = arr[0];
        for (int i = 1; i < n; i++) {
            prefixSum[i] = prefixSum[i - 1] + arr[i];
        }
    }

    public static void main(String[] args) {
        int[] arr = {10, 4, 16, 20};
        int n = arr.length;
        int[] prefixSum = new int[n];

        fillPrefixSum(arr, n, prefixSum);

        // Print the prefix sum array
        for (int i = 0; i < n; i++) {
            System.out.print(prefixSum[i] + " ");
        }
    }
}
```

The output of this code will be:

```
10 14 30 50
```

In this example, we've computed the prefix sum array for the input array `{10, 4, 16, 20}`. The prefix sum at index `i` is the sum of all elements from index 0 to `i` in the original array.

Use Cases

Prefix sums are particularly useful for solving problems related to range queries. For instance:

- **Sum of an array between indexes L and R:** Given an array `arr` of size `N`, and queries asking for the sum of elements from index `L` to `R`, you can use the prefix sum array to compute this efficiently.

Feel free to explore more applications of prefix sums! If you have any other questions or need further examples, feel free to ask. 😊

Remember, prefix sums are like the Swiss Army knife of algorithms—they come in handy in various scenarios! 🚀